

Aryan Kapoor

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Professional Summary

Computer Science student focused on cloud infrastructure, backend engineering, and DevOps. Hands-on experience building AWS architectures, Infrastructure as Code, event-driven systems, CI/CD pipelines, and containerized backend applications using Python and Flask. Experienced with cloud automation, asynchronous processing, microservices, and infrastructure analysis.

Professional Experience

Springer Capital

Aug 2025 – Nov 2025

Backend Intern

Remote

- Developed and debugged Python/Flask services within an 8-service microservice architecture using REST APIs, SQLAlchemy, and data-ingestion workflows
- Built mock API services and supported synchronous and asynchronous data-extraction workflows for development and integration testing
- Containerized backend services using Docker and Docker Compose, resolving networking, dependency, database initialization, and configuration issues
- Authored technical documentation covering API contracts, database schemas, service integrations, and architecture across the 8-service backend platform

Technical Projects

Project Hydra – Self-Healing Cloud Architecture

AWS | Terraform | SQS | EC2

- Built decoupled web and worker tiers using EC2 Auto Scaling, Application Load Balancer, S3, SQS, IAM, and VPC
- Designed asynchronous producer-consumer processing with Amazon SQS to separate web requests from background workloads
- Automated infrastructure provisioning with Terraform and implemented Auto Scaling for fault recovery and workload-based scaling

Helix – Serverless CDC Data Lakehouse

AWS | Lambda | DynamoDB | Athena

- Built an event-driven CDC pipeline using DynamoDB Streams and Lambda to transform transactional data into an analytics-ready S3 data lake
- Converted streaming records into partitioned Parquet datasets for serverless analytics using Amazon Athena
- Provisioned infrastructure with Terraform and developed Python data-transformation workflows using AWS SDK for Pandas

CloudOps – Infrastructure Intelligence Platform

AWS | Python | FastAPI | Boto3

- Built a platform that discovers AWS resources, models infrastructure relationships as graphs, and identifies potential security attack paths
- Developed resource discovery, infrastructure transformation, relationship analysis, and risk-scoring workflows using FastAPI and Boto3
- Created interactive infrastructure visualizations with Cytoscape.js and integrated EC2, S3, CloudFront, and IAM for deployment

Technical Skills

Programming Languages: Python, Java, C, C++, Bash, TypeScript

Cloud Platforms & Services: AWS, GCP, EC2, S3, Lambda, SQS, RDS, DynamoDB, VPC, ALB, Auto Scaling, CloudFront, IAM, Athena

DevOps & Infrastructure: Terraform, Docker, Docker Compose, GitHub Actions, CI/CD, Linux, Git

Backend Development: Flask, FastAPI, SQLAlchemy, REST APIs

Engineering Concepts: Microservices, Infrastructure as Code, High Availability, Event-Driven Architecture, Asynchronous Processing

Soft Skills

Technical Communication, Problem Solving, Analytical Thinking, Collaboration, Adaptability, System Design Thinking

Education

HKBK College of Engineering

2023 – 2027

Bachelor of Engineering in Computer Science

Bangalore, India

Vibgyor High

2019 – 2021

High School Diploma

Bangalore, India